Solution

(i) $\int 5x^4 dx = 5\int x^4 dx = 5\frac{x^{4+1}}{4+1} = 5\frac{x^5}{5} = x^5 + c.$

(ii) $\int\left(5x^2 - 4 + \frac{7}{x} + \frac{2}{\sqrt{x}}\right)dx = 5\int x^2 dx - 4\int dx + 7\int \frac{1}{x}dx + 2\int \frac{1}{\sqrt{x}}dx$

$= 5\frac{x^{2+1}}{2+1} - 4x + 7\log|x| + 2\frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + c$

$= \frac{5}{3}x^3 - 4x + 7\log|x| + 4\sqrt{x} + c$

(iii) $\int(2\cos x - 4\sin x + 5\sec^2 x + \operatorname{cosec}^2 x)dx$

$= 2\int \cos x\,dx - 4\int \sin x\,dx + 5\int \sec^2 x\,dx + \int \operatorname{cosec}^2 x\,dx$

$= 2\sin x + 4\cos x + 5\tan x - \cot x + c$

Example 11.9

Evaluate the following integrals:

(i) $\frac{12}{(4x-5)^3} + \frac{6}{3x+2} + 16e^{4x+3}$ (ii) $\frac{15}{\sqrt{5x-4}} - 8\cot(4x+2)\operatorname{cosec}(4x+2)$

Solution

(i) $\int\left(\frac{12}{(4x-5)^3} + \frac{6}{3x+2} + 16e^{4x+3}\right)dx$

$= 12\int \frac{1}{(4x-5)^3}dx + 6\int \frac{1}{3x+2}dx + 16\int e^{4x+3}dx$

$= 12\left(\frac{1}{4}\right)\left(-\frac{1}{2(4x-5)^2}\right) + 6\left(\frac{1}{3}\right)\log|3x+2| + 16\left(\frac{1}{4}\right)e^{4x+3} + c$

$= -\frac{3}{2(4x-5)^2} + 2\log|3x+2| + 4e^{4x+3} + c.$

(ii) $\int\left(\frac{15}{\sqrt{5x-4}} - 8\cot(4x+2)\operatorname{cosec}(4x+2)\right)dx$

$= 15\int \frac{1}{\sqrt{5x-4}}dx - 8\int \cot(4x+2)\operatorname{cosec}(4x+2)\,dx$

$= 15\left(\frac{1}{5}\right)(2\sqrt{5x-4}) - 8\left(\frac{1}{4}\right)(-\operatorname{cosec}(4x+2)) + c$

$= 6\sqrt{5x-4} + 2\operatorname{cosec}(4x+2) + c.$

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